

Overview of Intelligent Textbooks in 35 Minutes

Democratizing Education for Using Intelligent Textbooks

- AI in Education Conference
- St. Mary's University
- September 23rd, 2026
- Dan McCreary



READ THIS FIRST

A PDF is a Poor Representation of a Set of Interactive MicroSimulations!

These slides were built to be *used*, not just read. 8 of them embed live MicroSims — small simulations you explore with sliders, buttons and hover. In this PDF each one is frozen into a static snapshot.

View the full interactive slide deck:

[https://dmccreary.github.io/intelligent-textbooks/slides/slide-viewer.html?
input=ibooks-in-35-minutes/index.md](https://dmccreary.github.io/intelligent-textbooks/slides/slide-viewer.html?input=ibooks-in-35-minutes/index.md)

Scan the QR code, or click any MicroSim snapshot in this PDF to open the live version.

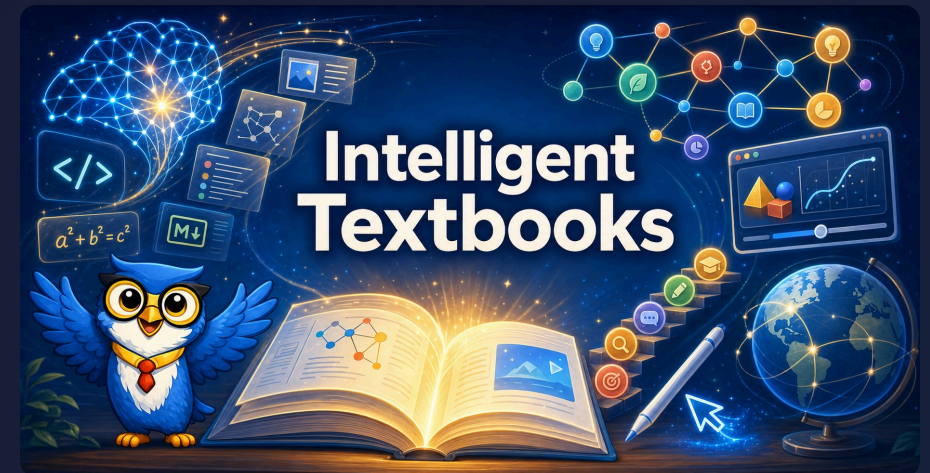
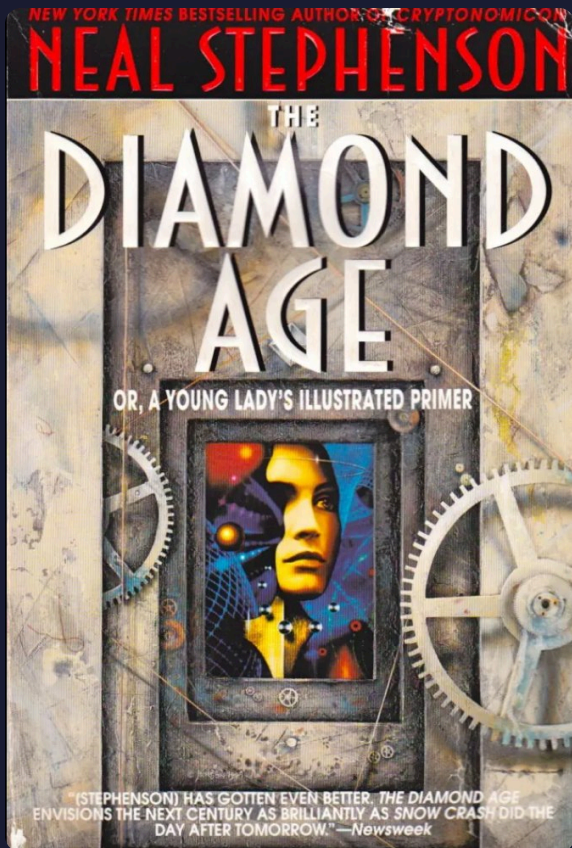


Scan for the interactive slide deck

My Original Inspiration

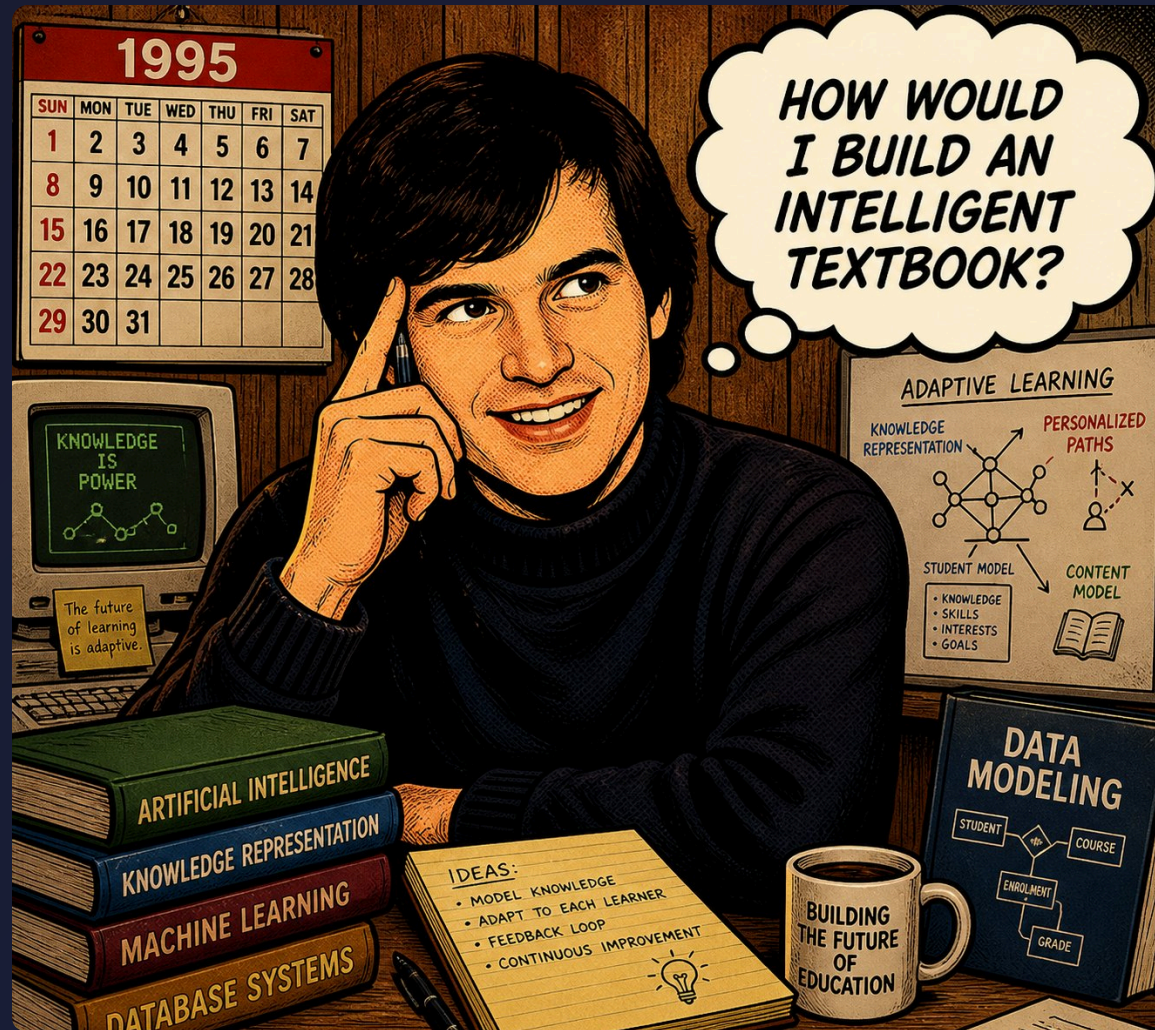
Diamond Age: Neal Stephenson's 1995 Cyberpunk Novel

- A young girl is given an AI-powered tablet that imprints on the girl
- Every lesson the girl needs is customized to her context



Dan's Quest

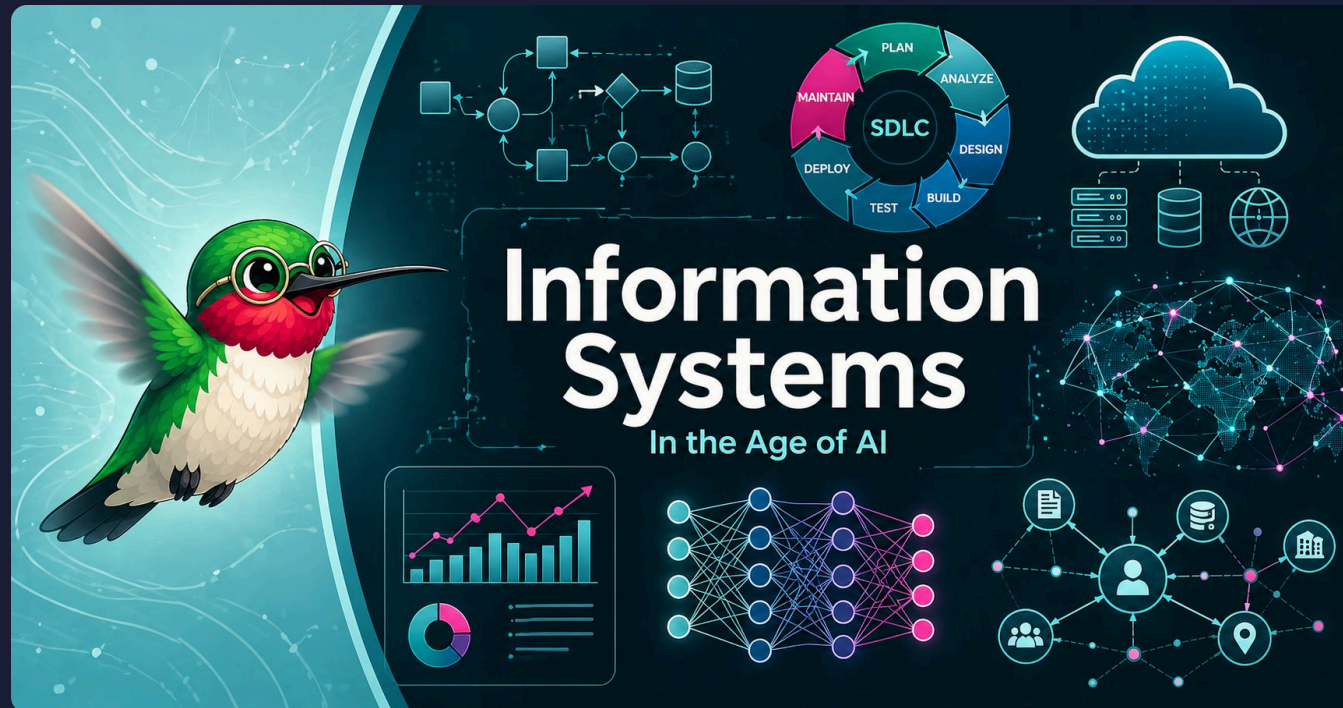
- How would I build such a device?



The Problem With Paper Textbooks

- The static textbook problem (printed in 2019, used in 2026)
- One-size-fits-all "Direct Instruction" model fails the 80% - the advanced bored, the struggling get lost
- Cost & access gap: US college students pay \$1,300 per year on out-of-date printed textbooks
- The engagement crisis (passive reading vs. active learning)
- Teachers are drowning (no time to personalize)

Prompt Goal: *"Generate a 1,000 page textbook on information systems with 100 MicroSims"*



Current Status

What you can do now (available since October 2025):

"Hey Claude, please generate a high quality interactive intelligent textbook.

Use Dan McCreary's intelligent textbook skills <https://github.com/dmccreary/ibook-skills>

Use the following course description...PATH TO YOUR COURSE DESCRIPTION...

Create a Learning Graph with 500 Concepts".

What is an Interactive Intelligent Textbook

- A textbook with "a brain" (learning graph) that can be classified into five levels
- A simple but powerful network of concepts and their learning order dependencies
- A simple classification system for concepts (taxonomy)
- Forced interactivity (few plan-old-images)
- No reliance on bandwidth hungry videos
- Small lightweight simulations (500 lines of JavaScript in an 10K byte download)
- Focus on structures that generate events that predict student mastery of a concept
- Hooks for easy migration to a xAPI/LRS infrastructure
- Designed to migrate from level 2.9 to level 5 without changing the data structures

Health Education

Hey Claude, Please generate age-appropriate six textbooks base in the Minnesota Department of Education guidelines:

Generated six textbooks which shared content.



Infographic Posters

Infographic Posters

Families: Many Ways To Care



Local Minnesota Participants

- The Thinking Spot - Monday "Office Hours"
 - **Rima Parikh** - Coding Club and STEM Kits
 - **Arun Batchu** - Testing Skills, Scratch
 - **Rick Tanler** - Understanding Dementia
- MicroSims
 - **Valarie Lockhart** - Originator of MicroSims
 - **Troy Peterson** - WikiTube
- ISD 197
 - **Miles Lawson** - Digital Citizen Ship
 - **Peter Olson-Skog**
- Groves Learning Organization
 - **Curtis Olufsin** - Automation, ASL
 - **Andy Tolin** - Coding Club
- My Mentees
 - **Anvith Pothula** - Graph Neural Network Textbook
 - **Rishik Kondadadi** - Seizure Awareness in Minnesota Schools
- Stone Arch Collaborative
 - **Shannon Seaver** - AI Leadership in schools - CreateMPLS
 - **Dr. Randy Smasal** - Change management consulting
 - **Lori Ryan** - textbooks for marketing and brand management
 - **Stephanie Lensing** - textbooks for marketing and brand management
- University of St. Thomas
 - **Dan Yarmoluk** - Digital Transformation
 - 12 Student Textbooks
- University of Minnesota
 - **Dr. Sharat Bhatra** - Circuits, Senior Seminar, 12 Students
 - **Dr. Jarvis Haupt** - Signal Processing
 - **Dr. Gan Qui** - Physics of Semiconductor Devices

What It Will Generate

1. Extended Course Description - enriched with Learning Objectives using the Bloom Taxonomy and Quality Score
2. Inferred Concept List - between prerequisites and objectives
3. Learning Graph (Concept Dependency Graph) - the **ground truth** of the textbook
4. Book Mascot Character Sheet - enhanced engagement
5. Book Chapter Design - Focus on coverage and balance
6. Book Chapter Content - heart of the textbook
7. Placeholders for diagrams, figures, drawings, infographics, simulations - all interactive
8. Microsims - interactive elements that tracks mouse events (hover and clicks)
9. Glossary of Terms - precise, concise, distinct definitions with examples and links
10. FAQ - common questions and answers with links
11. Quizzes - multiple choice
12. References - with annotations for relevance
13. Graphic Novel Stories
14. Book Type Specific Content - code executions, circuit simulations, timelines, maps, pronunciation, flashcards, lesson plans, challenge cards, kit book covers
15. Custom Cover Page - social media previews
16. About this Book
17. Landing Page with cover
18. Appendices, Contact, License, Teachers and Instructor's Guides
19. Book statics - quality reports, book metrics, reading level checks
20. Book announcements for social media and README.md

MicroSims

- Worked as a volunteer with a local coding club starting in 2014
- Worked with Valarie Lockhart on teaching prompt engineering to teachers
- Valarie Lockhart coined the term "MicroSim" after using ChatGPT to generate p5.js
- I generalized the process and used iframes to make MicroSims easy to embed
- Published a paper formalizing the MicroSim standards with Valarie and Troy Peterson

Monday, October 13, 2014

Motors for Arduino Labs

This week I went through many of the SparkFun Inventor's Kits that we use in our CoderDojo Arduino labs and realized that the kids have been loving our little DC mini-motors to death. As you can see in Figure 1 over half of the wires had broken off!

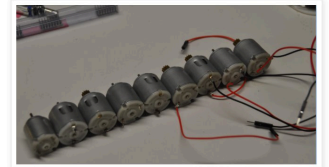
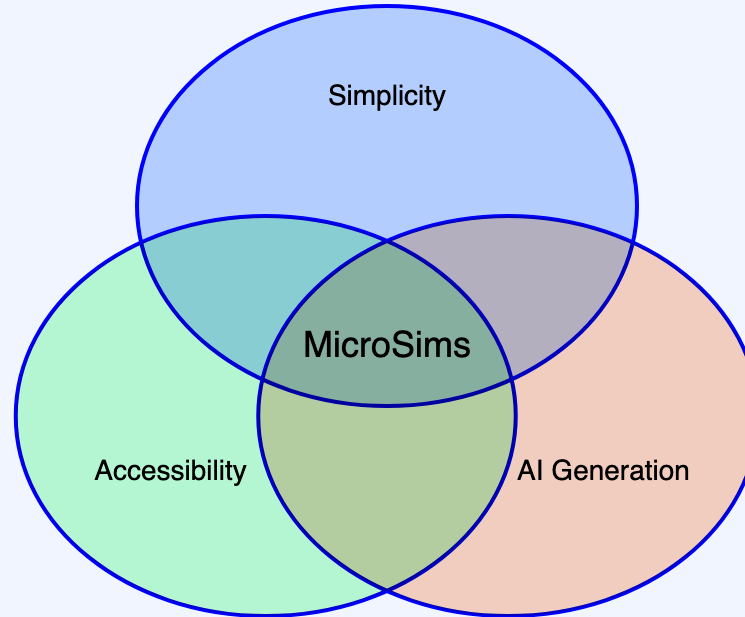


Figure 1: DC Motors with most connector broken off

MicroSim Uniqueness

What Makes MicroSims Unique



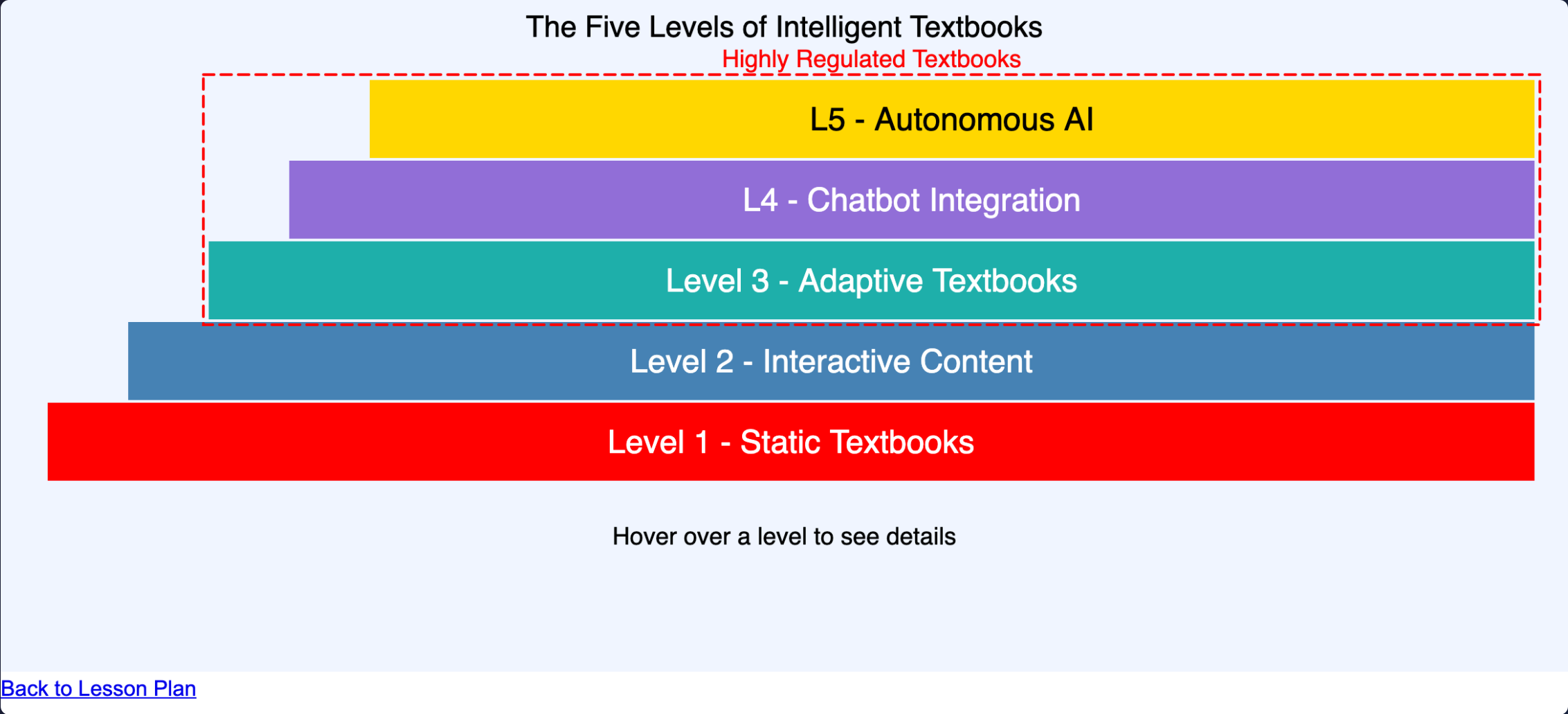
Hover over regions to learn more

Click on any region to learn more about what makes MicroSims unique

Static snapshot — click to run this MicroSim live

[MicroSim Uniqueness](#)

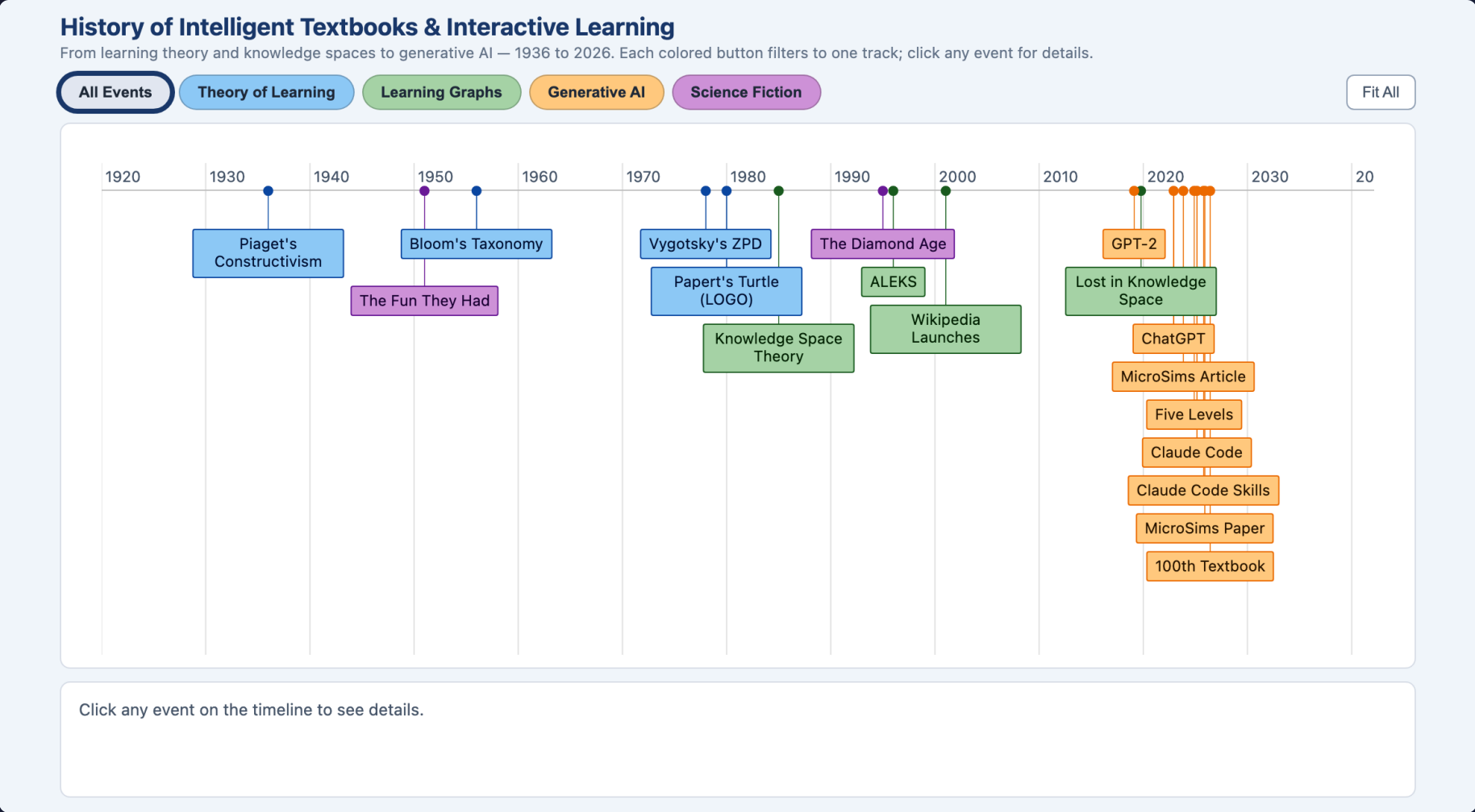
The Five Levels of Intelligent Textbooks



Static snapshot — click to run this MicroSim live

[Book Levels Fullscreen](#)

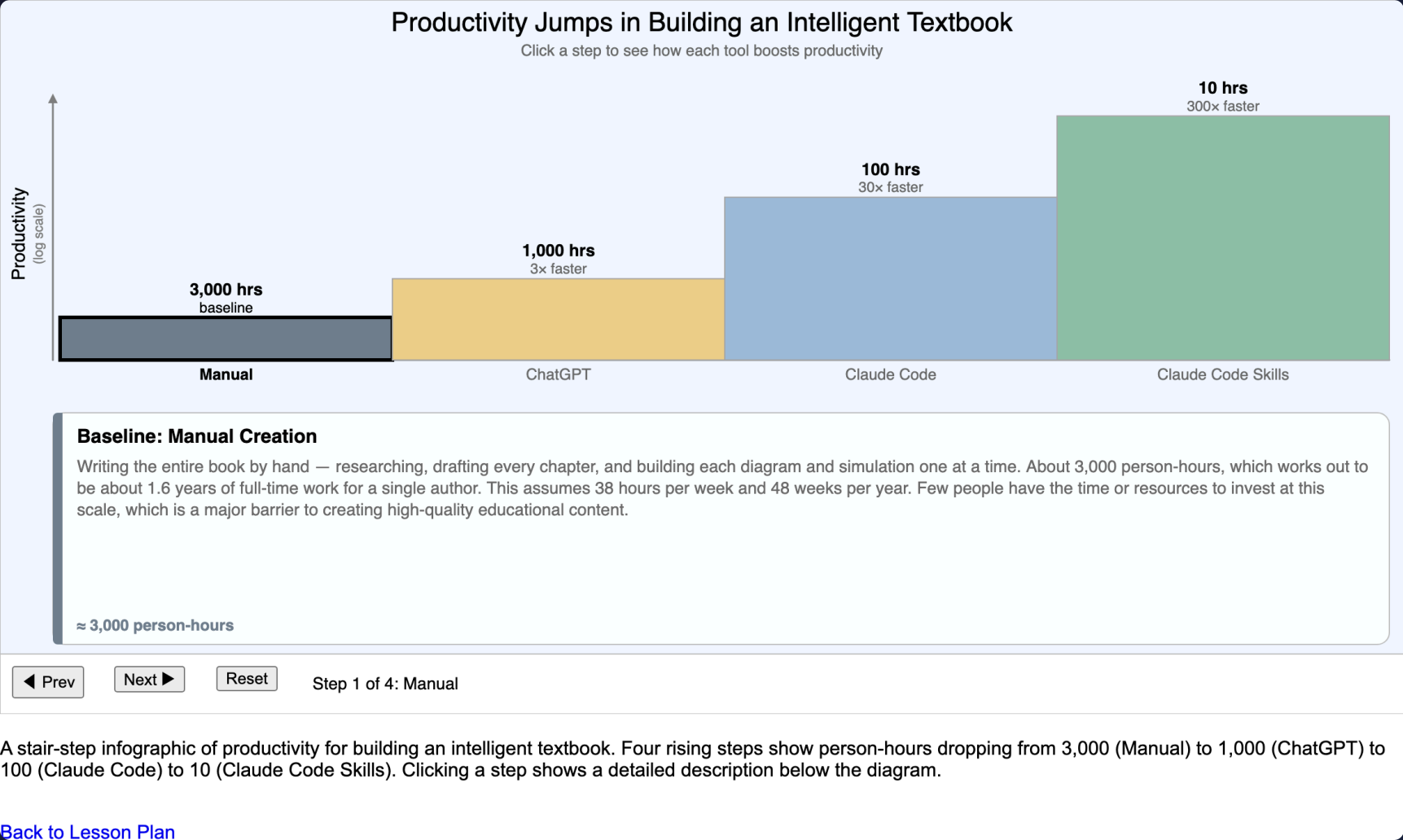
Timeline of Intelligent Textbooks



Static snapshot — click to run this MicroSim live

[Timelines Fullscreen](#)

Productivity Jumps



Static snapshot — click to run this MicroSim live

Productivity Jumps Fullscreen

Final 10 Hours

- Most time-consuming tasks was user interface layout cleanup of the MicroSims (placement)
- New Claude Vision tool now "sees" UI and will apply layout fixes
- Saves and additional 3-4 hours per book
- Automation of mascot design and cover design also saves 1 hour

Current Status (this is real now)

- 120 level 2.99 Books
- Getting ready to go to level 3 (cost effectively) using xAPI and LRS
- 6,000 MicroSims
- 350+ Mini Graphic Novels
- 300+ Interactive Infographic Posters
- Ongoing research on [Automating Instructional Design](#)
- Continued integration of [The Learning Sciences](#)

[Case Studies](#)

Target Classroom

The target classroom is a remote school in Africa that has limited internet bandwidth. Our goal is to give each of these students the same MIT-caliber education as the most advanced colleges and universities in the US today.



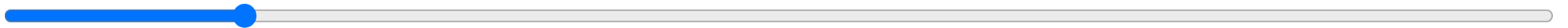
The Bouncing Ball MicroSim Example

Bouncing Ball Simulation



Start

Speed: 3

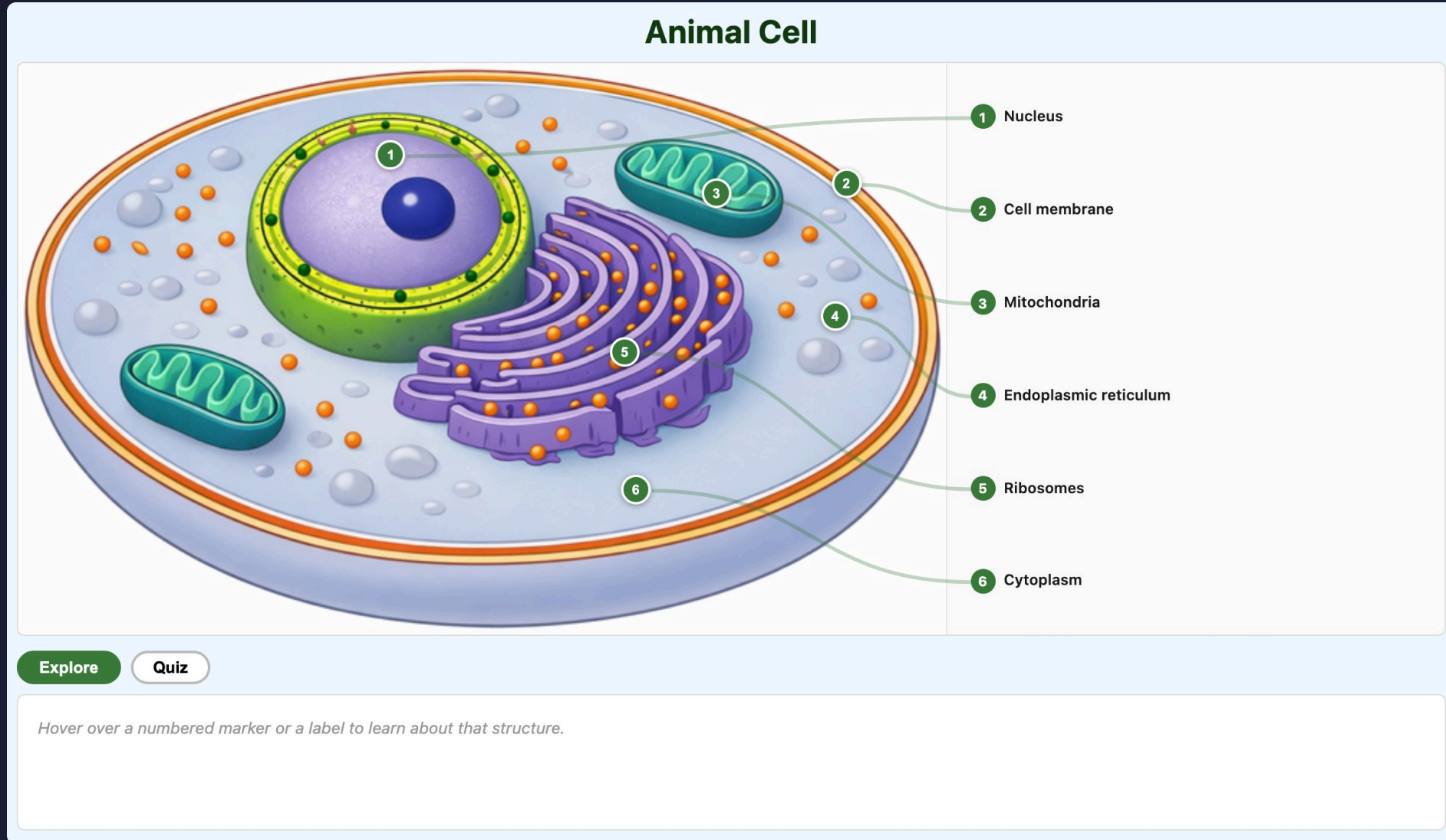


Interactive bouncing ball simulation with speed control and start/pause button.

[Back to Bouncing Ball Lesson Plan](#)

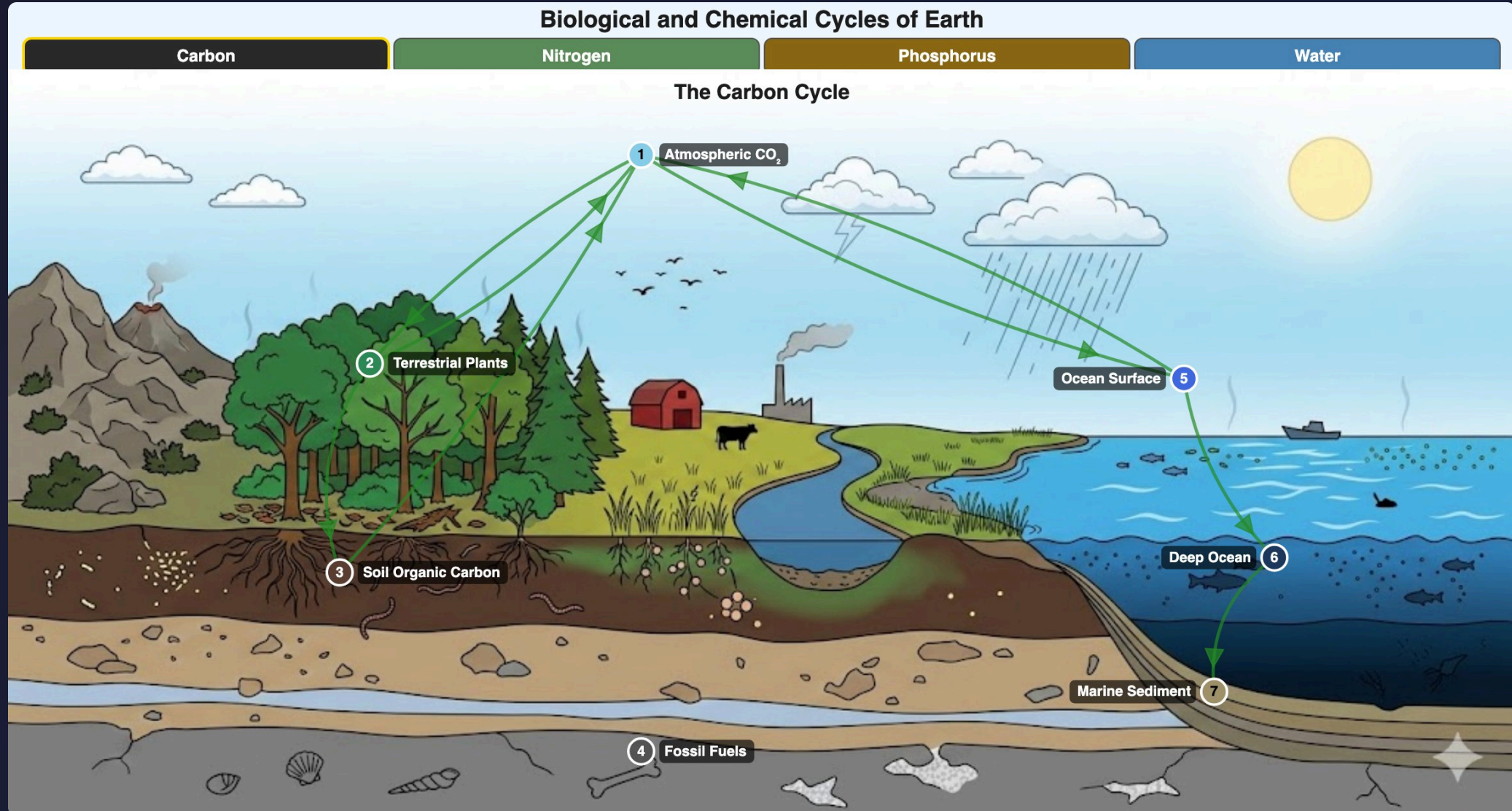
Static snapshot — click to run this MicroSim live

Animal Cell (Interactive Infographic Overlay)



Static snapshot — click to run this MicroSim live

Biogeochemical Cycles (Water, Nitrogen, Carbon, Phosphorus)



Static snapshot — [click to run this MicroSim live](#)

Challenges Getting LLMs to Generate Consistent Content

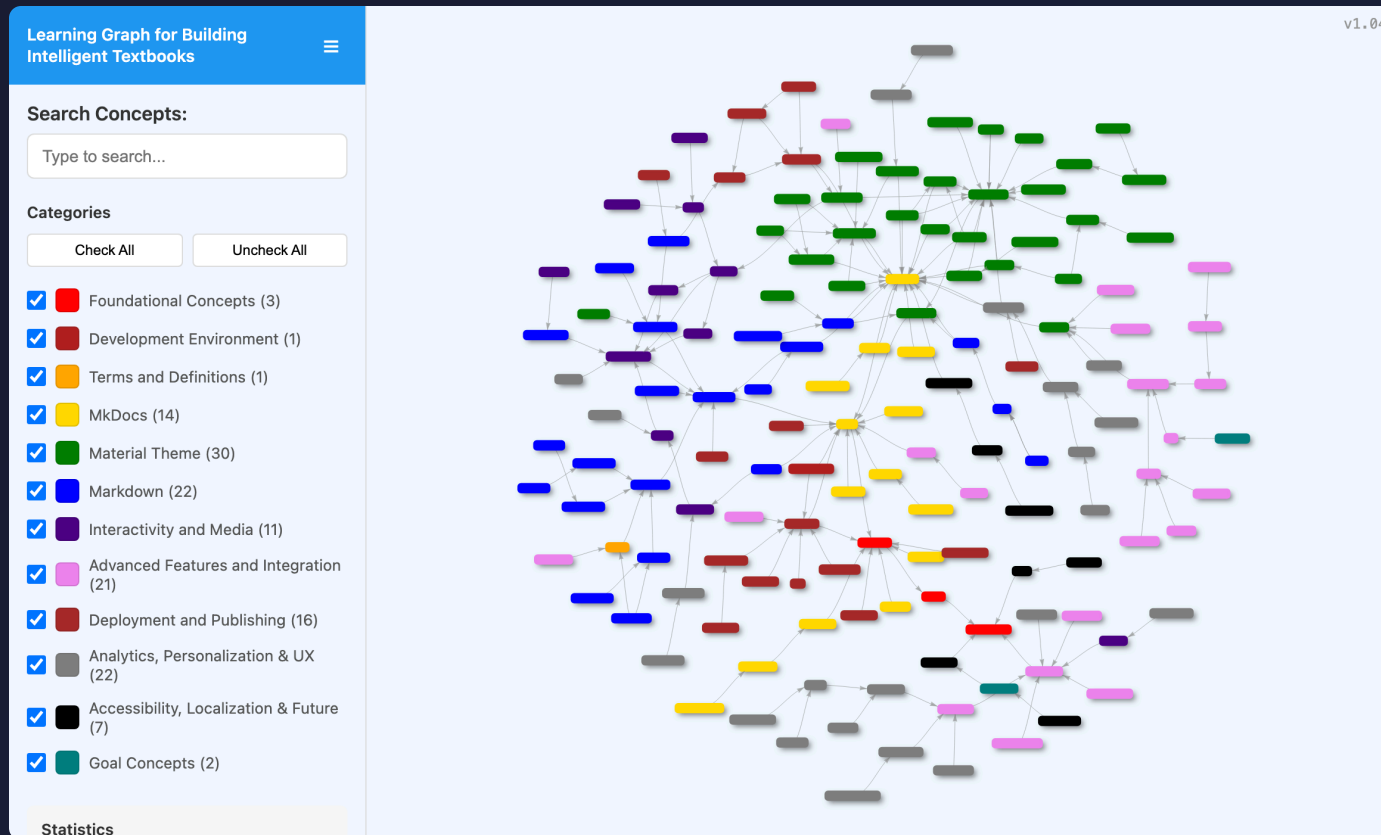
- Getting LLMs to generate consistent simulation interfaces
- Every diagram MUST be interactive (for level 3 xAPI enabled)
- No static images or drawing
- Complex infographics are great, but overlay interactive region hovers

How I Generate Intelligent Textbooks

1. Course description (100 point scale)
2. => Learning graph
3. => Chapter design
4. => Chapter content generation (with mascot and microsim specifications)
5. => MicroSim generation
6. => Supplementary content generations (glossary, faq, quizzes, lesson plans, references)
7. => Automated quality assessment (coverage, metrics, scaffolding)

Learning Graph

- Core data structure behind content generation and hyper-personalization.
- Directed Acyclic Graph of concepts and their learning order



Static snapshot — click to run this MicroSim live

[Computer Science \(400\)](#), [Information Systems \(580\)](#)

Book Mascots



Book Mascot Montage

Sample Titles

Digital Citizenship (ISD 197)

Personal Finance (State Guidelines)

Learning Python

Scratch

Learning Linux

Coding Clubs

Learning MicroPython (Physical Computing)

AP Courses: Calculus, Physics, Chemistry, Biology, Natural Sciences

Algebra

Data Science

Deep Learning

Signal Processing (University of Minnesota)

IB Courses: Theory of Knowledge, Functions

Ecology (AP Natural Sciences)

Quantum Computing (a skeptics guide)

Dementia (with Rick Tanler)

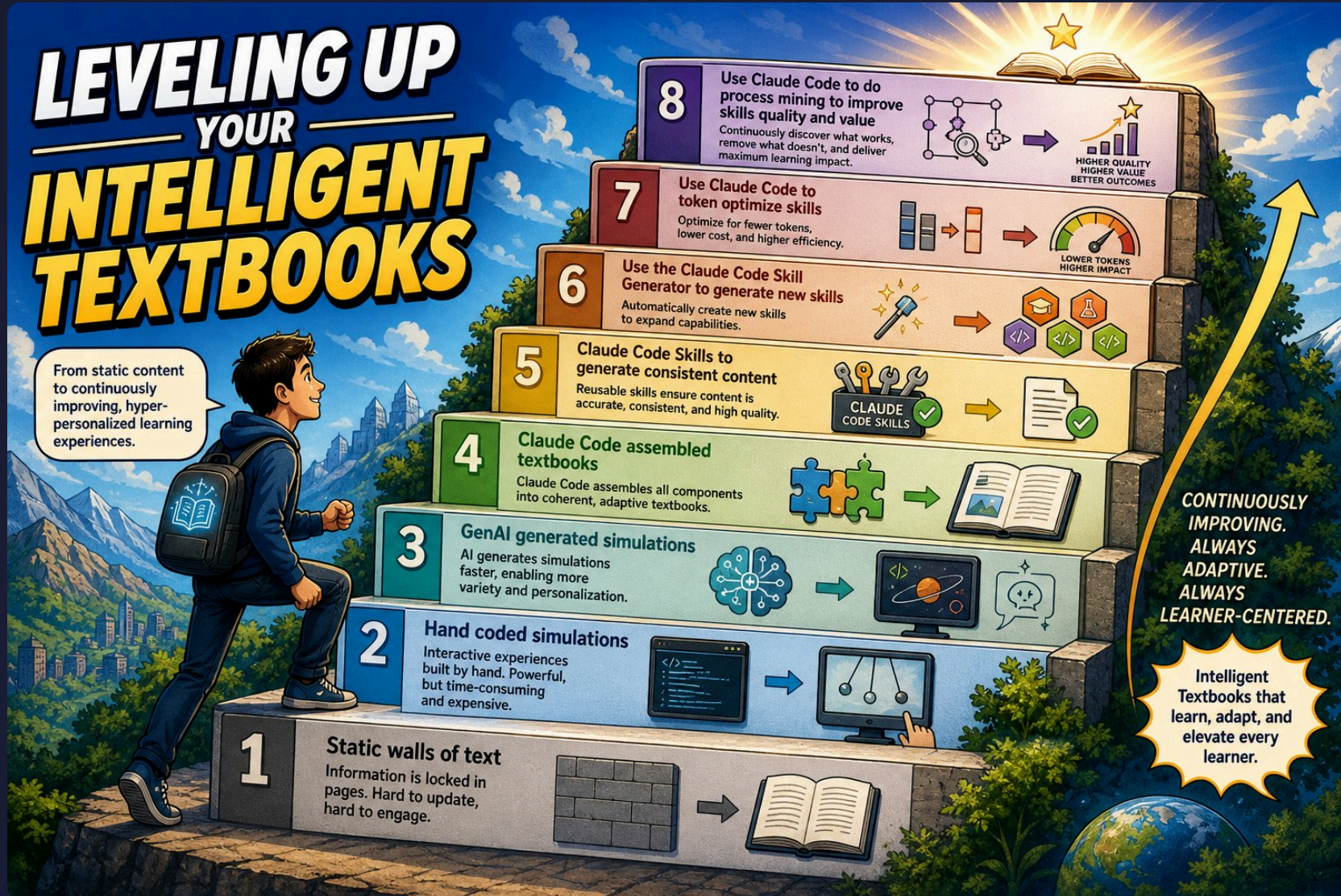
Digital Transformation (with Daniel Yarmoluk)

Cooking Science

Public Health

Forensic Science

Leveling Up Your Intelligent Textbooks



Leveling Up Your Intelligent Textbooks Text

1. Static walls of text
2. second level "Hand coded simulations"
3. GenAI generated simulations
4. Claude Code assembled textbooks in GitHub
5. Claude Code Skills to generate consistent content
6. Use the Claude Code Skill Generator to generate new skills
7. Use Claude Code to token optimize skills
8. Use Claude Code to do process mining to improve skills quality and value

Skills for Developing Intelligent Textbooks

1. Created with Claude Code starting around October 2025
2. Currently about 13 core skills
3. Thoroughly tested with Claude Code works on OpenAI ChatGPT and Google Gemini
4. Continually being optimized for token efficiency – targeting \$2/textbook

The Democratization of Education

- A Side Effect of Dan's Quest
- How would the world change if a 12-year old girl from the remote regions of Africa has access to the **same quality** of hyper-personalized education as every student at MIT?
- How could we use intelligent textbooks to democratize education for all children on Earth?
- Should an ultra-high quality of hyper-personalized education be free for all children on earth?

The Democratization of Education

Education is a fundamental human right. Now, technology makes it possible for every child on Earth to learn, grow, and thrive.

The DEMOCRATIZATION of EDUCATION

Every child. Everywhere. Unlimited potential.

ONE PLANET. ONE FUTURE. EDUCATION FOR ALL.

EQUAL ACCESS
Every child, everywhere, no exceptions.

100% FREE
High-quality education is a right, not a privilege.

INCLUSIVE
Designed for all learners, in every language and culture.

EMPOWERING
Unlocking potential and creating brighter futures.

QUALITY WITHOUT COMPROMISE
World-class content, built by experts, continuously improved.

BETTER OUTCOMES
Higher achievement, greater engagement, stronger futures.

STRONGER SOCIETIES
An educated world is a more prosperous, peaceful, and equitable world.

SUSTAINABLE IMPACT
Technology and collaboration create lasting change for generations to come.

POWERED BY
LEVEL-5
INTELLIGENT TEXTBOOKS

AI that understands. Content that adapts. Learning that's personal for every child.

HYPER-PERSONALIZED
Adapts to each learner's level, pace, strengths, and goals.

MASTERY-BASED
Ensures deep understanding before moving forward.

REAL-TIME FEEDBACK
Instant support and guidance when learners need it most.

TUTORING ON DEMAND
AI tutor available 24/7 in any language.

RICH & INTERACTIVE
Multimedia, simulations, and real-world experiences bring learning to life.

SAFE & PRIVATE
Built with privacy, security, and child safety at the core.

THE FUTURE IS OPEN. THE FUTURE IS FOR ALL.
LET'S BUILD IT TOGETHER.

When we educate every child, we elevate humanity.

A smarter world starts with an educated world.

Making Teaching Fun

- From "Sage on the Stage" to Guide on the Side

Making Chatbot 41x lower Cost

Benchmarking Knowledge Retrieval Architectures Across Educational and Commercial Domains: RAG, GraphRAG, and Compact Knowledge Graphs

Daniel Yarmoluk*

Dan McCreary†

Version 0.6.2 — April 19, 2026

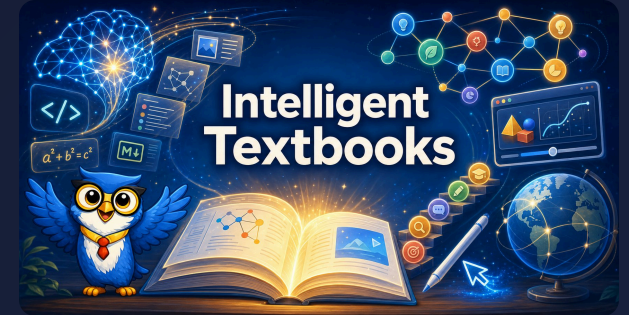
Abstract

Retrieval-augmented generation (RAG) and graph-based retrieval (GraphRAG) are the dominant paradigms for grounding LLM responses in structured knowledge. Both optimize for recall while treating token cost as a secondary concern. We introduce Compact Knowledge Graphs (CKG)—pre-structured DAG representations with explicit concept taxonomy and pipe-delimited dependency encoding—and present a two-track benchmark comparing CKG, RAG, and GraphRAG across educational and commercial domains.

Call to Action

- Try it out yourself!
- Requires a \$20/month Claude Pro, ChatGPT or Google Gemini account
- Use Claude Sonnet for 95% of skills
- Generate 6-8 textbooks per month (in three 5 hour windows per day)
- Per textbook cost \$2-\$3 depending on complexity of MicroSims

Because every child on Earth can be Nell



Appendix

[WikiTube Solar System Simulation](#)